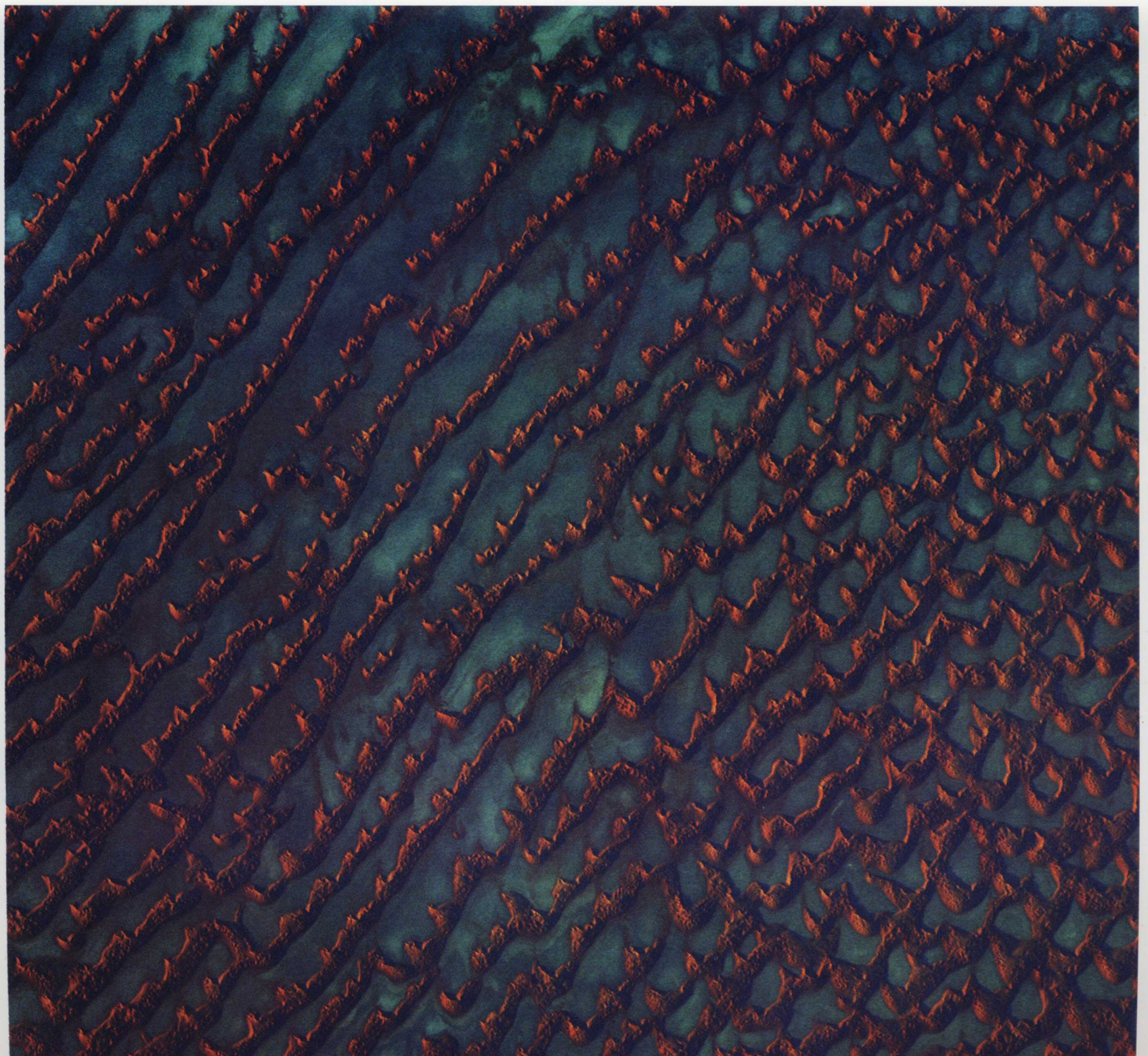
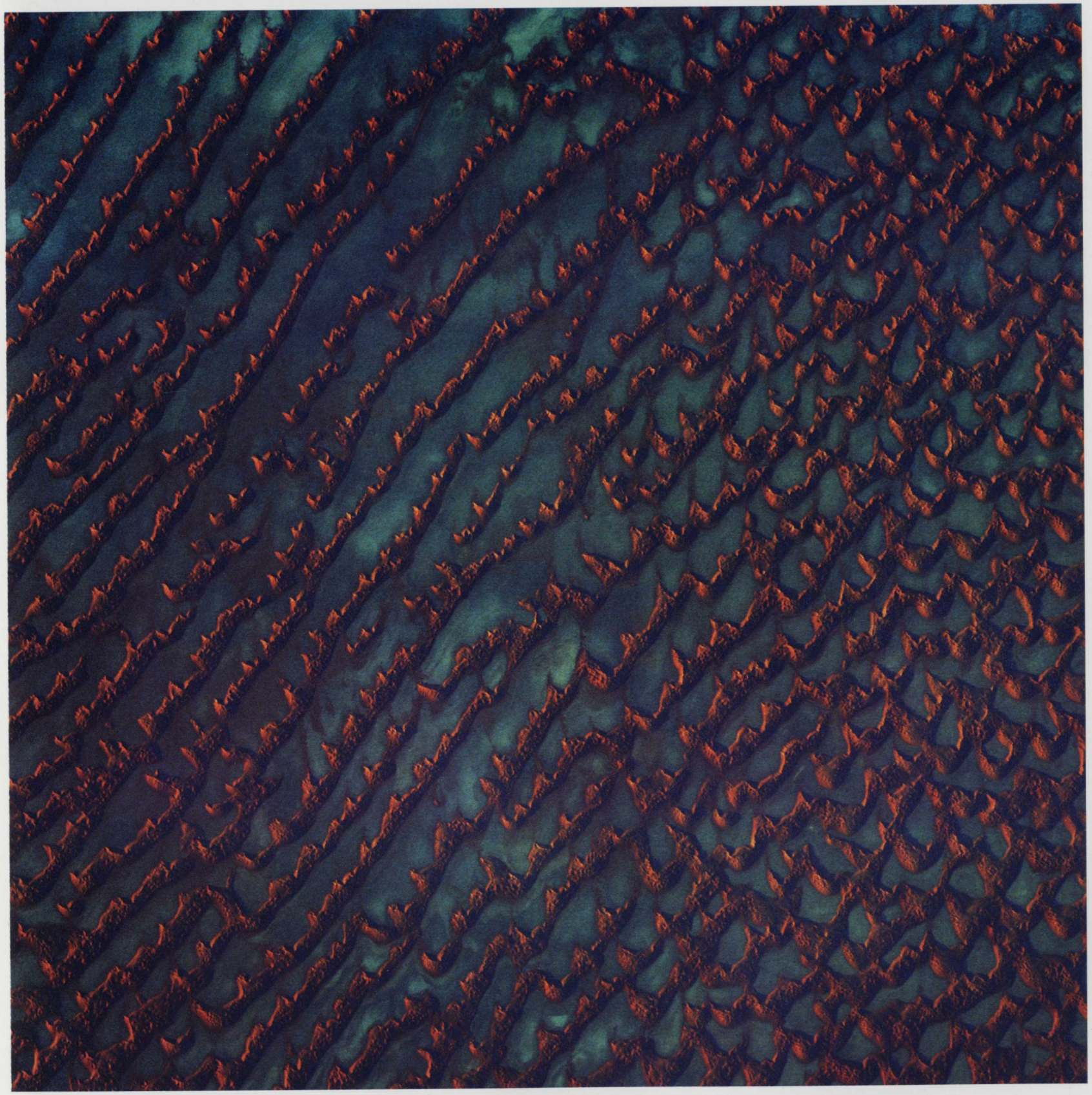
NATIONAL GEOGRAPHIC SOCIETY

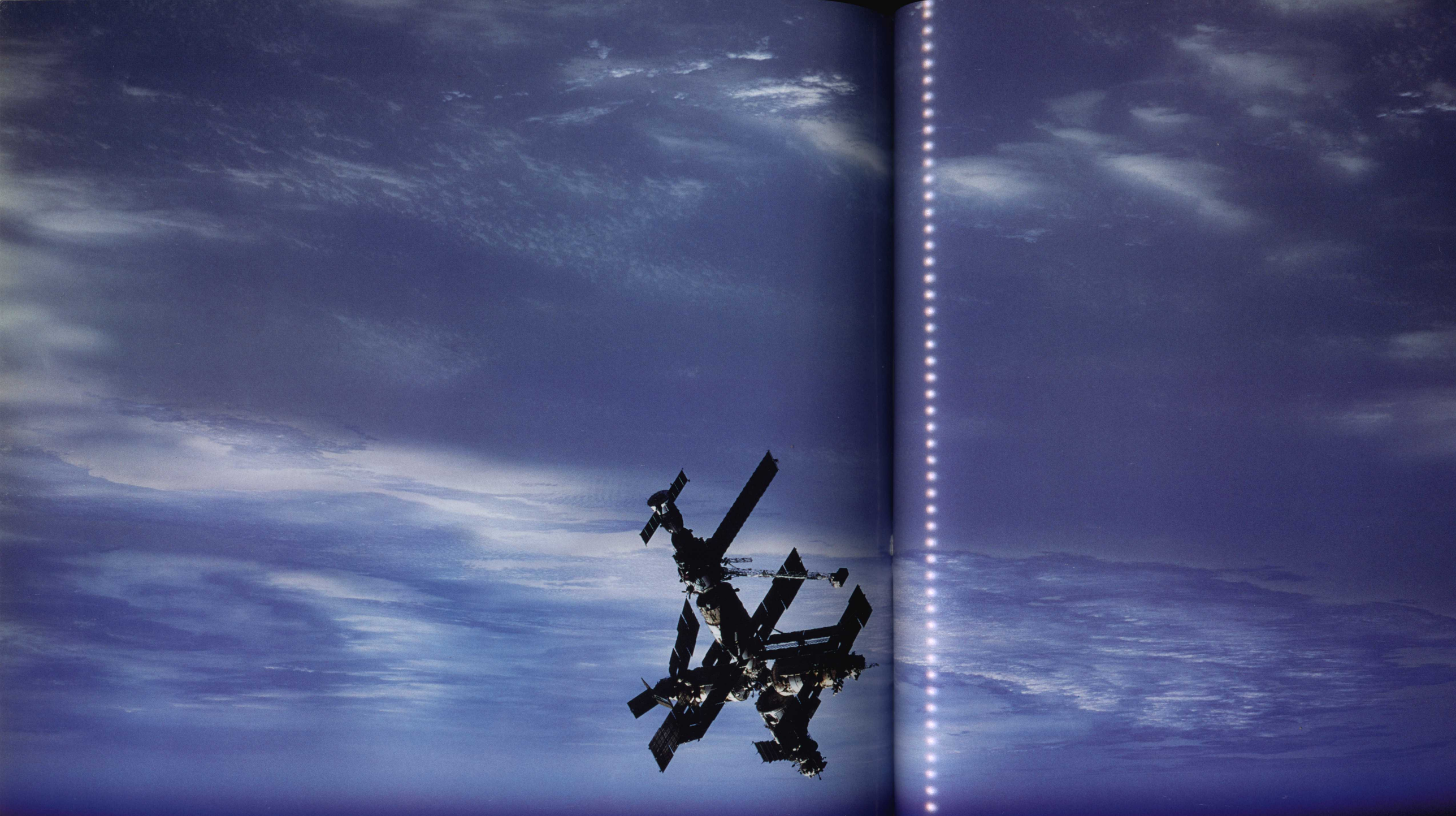
NASA ASTRONAUTS PHOTOGRAPH THE EARTH

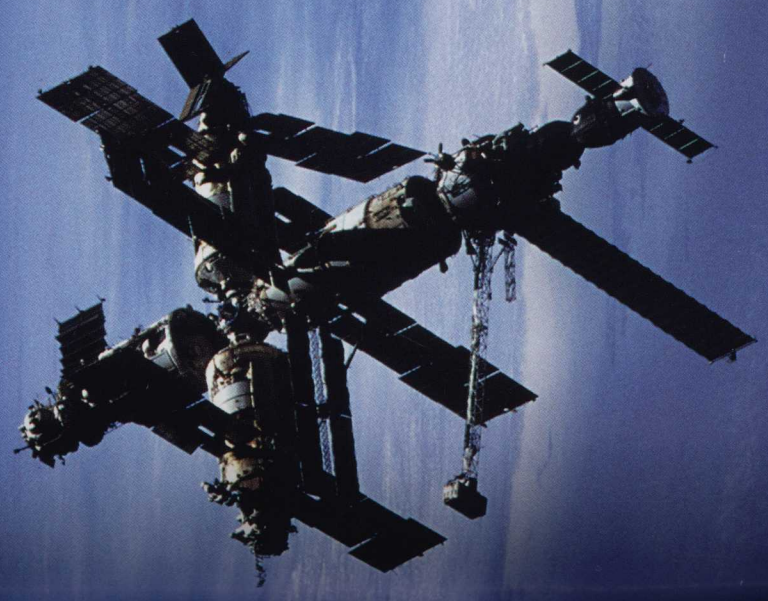
ORBIT

Jay Apt, Michael Helfert, Justin Wilkinson



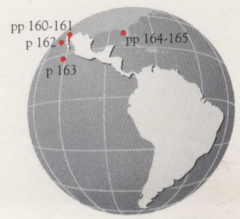






Mouth of the Colorado

The motions of Earth's crust have forced the U.S. Southwest up, and the land cracked as it expanded. It bleaches in the sun except where it is watered by the mighty Colorado. The river (flowing into the picture from the left) irrigates farmlands in the United States, and empties into the north end of the Gulf of California (right). Seven states from Wyoming to Arizona, as well as Mexico and certain Native American tribes, all have legal claims to some of the water supply of the Colorado River Basin. Only a salty trickle now flows all the way into the Gulf of California. In contrast to the yellow dune sands of the Altar Desert (center), the farmlands appear as green areas along the river and to the south of the 30-mile-long Salton Sea, the pear-shaped lake at the lower left.



Los Angeles and San Diego

The weather of the Los Angeles basin is controlled by the cool California Current and a high-pressure system offshore. The benefit is the stable climate of sunshine and little rain. A consequence is smog. The cool moist air pushed into the valley by the ocean winds is held in place by warm air descending from the high-pressure system. This warm layer on top of the cool marine air forms a natural lid—an inversion layer—that holds the air pollutants near the ground. These chemical emissions are trapped and thicken into smog (photo at right). Smog is not new: Spanish explorers saw Indian fires forming a light smog capped and held in place by an inversion layer in the 1500s. In this December 1983 scene, smog is wafting offshore from Point Vicente across Santa Catalina Island and out into the Pacific north of San Diego. Reduced emissions from factories and vehicles have lessened the severity of the smog in the last decade.

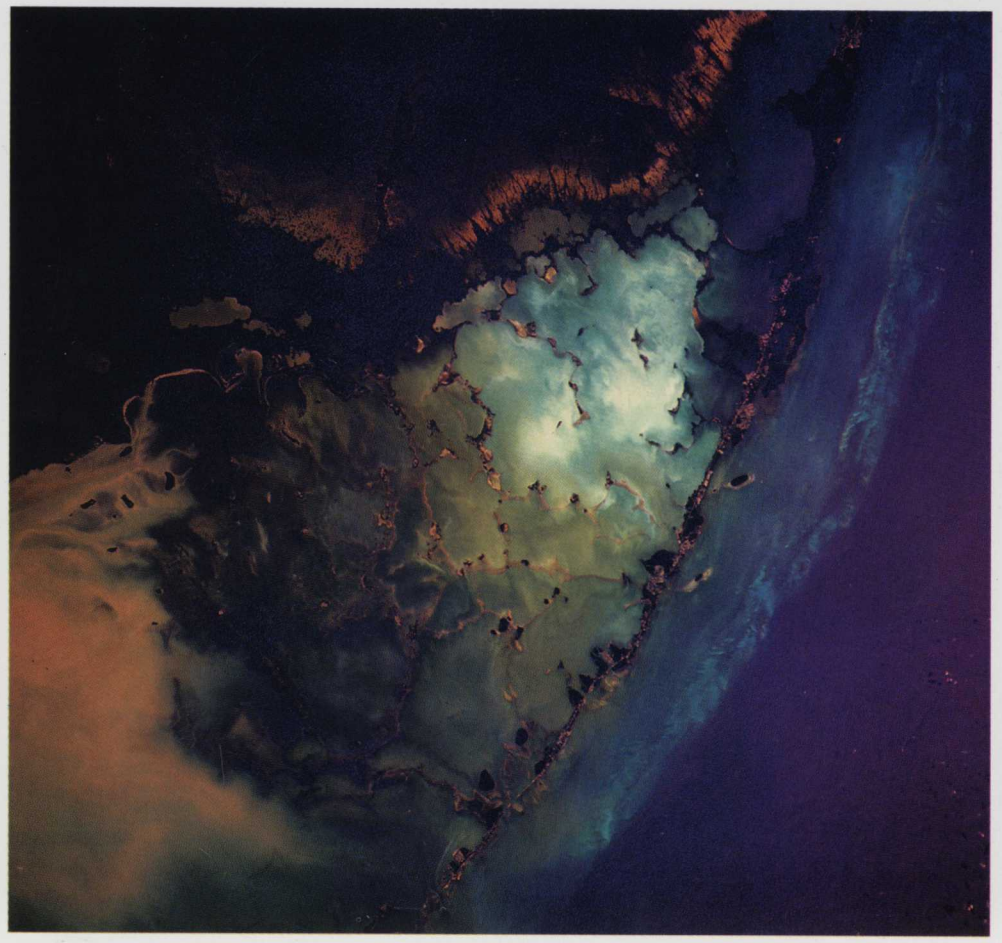
Florida from the North

LEFT

While scientists performed medical research in *Columbia's* laboratory, her pilot captured this view, from Cape Canaveral in the lower left southward to the curving horizon. Lake Okeechobee is the large cloud-free lake. Elsewhere, the sea breeze combines with moisture breathed into the air by the wetlands' water and vegetation to create cumulus clouds that water the southern half of the peninsula. Miami is the large city to the east (left) of Lake Okeechobee; the Everglades are to the south. On the west coast the port cities of Tampa and St. Petersburg are clustered around Tampa Bay. Ninety miles to the south of the Florida Keys is the narrow island of Cuba.

Florida Bay and the Keys

South of the Everglades, Florida Bay is frequently choked with brown algae; at other times it may be clear. Since 1987 the bay's sea grass habitat has drastically shrunk, the water has been murky, and pink shrimp populations have declined. The algae blooms seen here were fueled by nutrients released by dead and decaying sea grass. Because of water management practices in south Florida, less fresh water is flowing into the bay, making the bay saltier. This has killed the sea grass and allowed bottom sediments to be stirred by winds and currents.

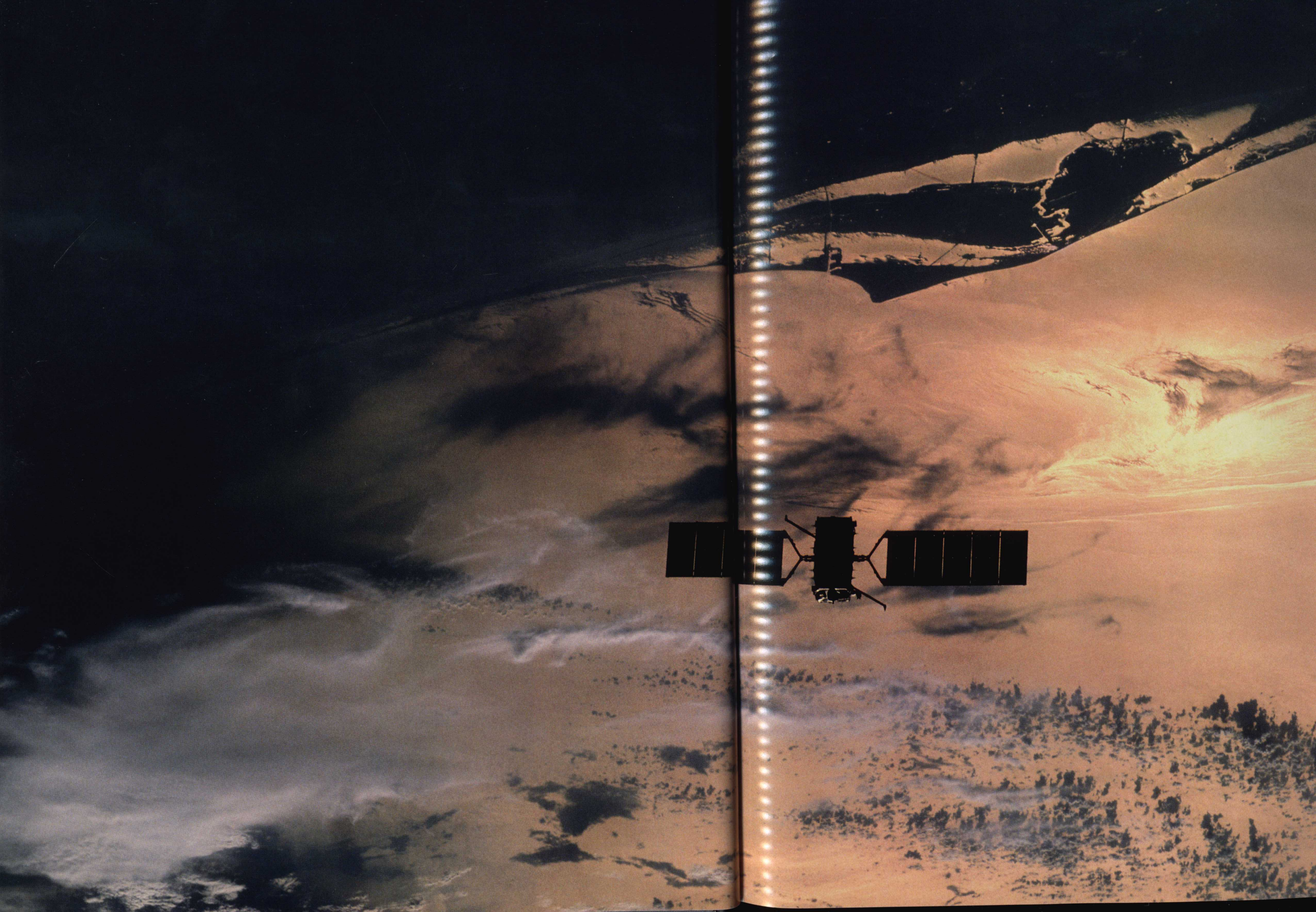


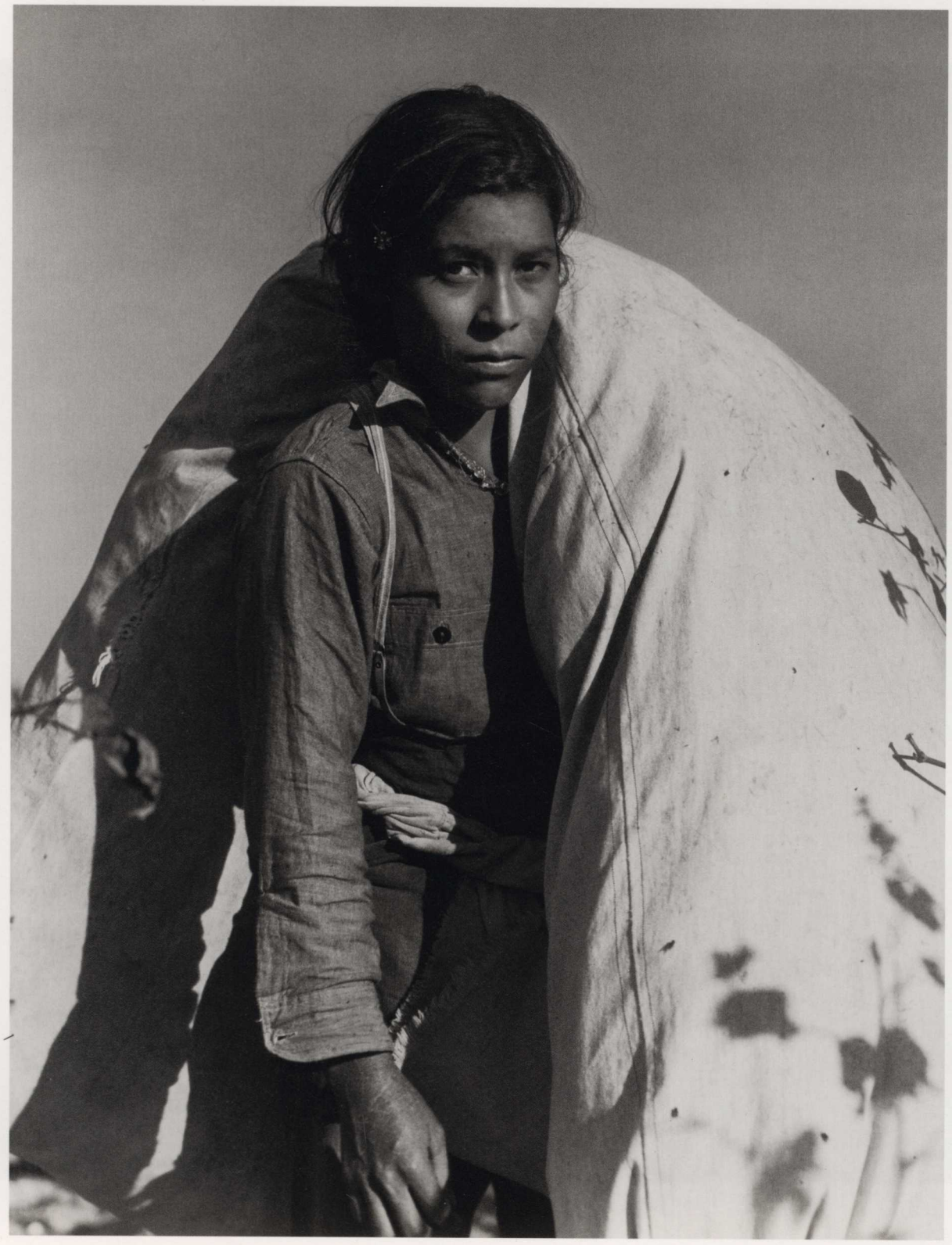
Cape Canaveral and the Eureca Satellite

FOLLOWING PAGES

Europe's first reusable spacecraft sails above Kennedy Space Center five hours after the crew of *Atlantis* released it. Ten months later, *Endeavour* left Earth from a launch pad just to the left of the prominent cape in the photo to retrieve the satellite. Whenever we sail into space from here, we expect to be called on to do whatever it takes to accomplish our mission. Two hours of hard work in space suits outside the Shuttle awaited *Endeavour's* crew, as they had to latch down Eureca's 66-foot-long solar panels by hand when robotic systems commanded from the ground failed during the flight.

When humans finally reach out to the stars, some of the crew will leave Earth from here.





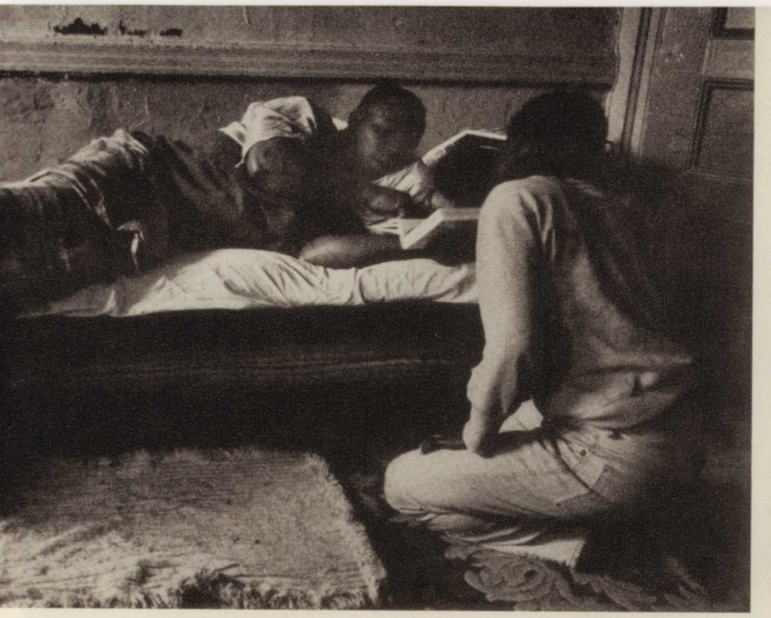
91 *Young Cotton Picker, San Joaquin Valley, California*, November 1936

GORDON PARKS

AR

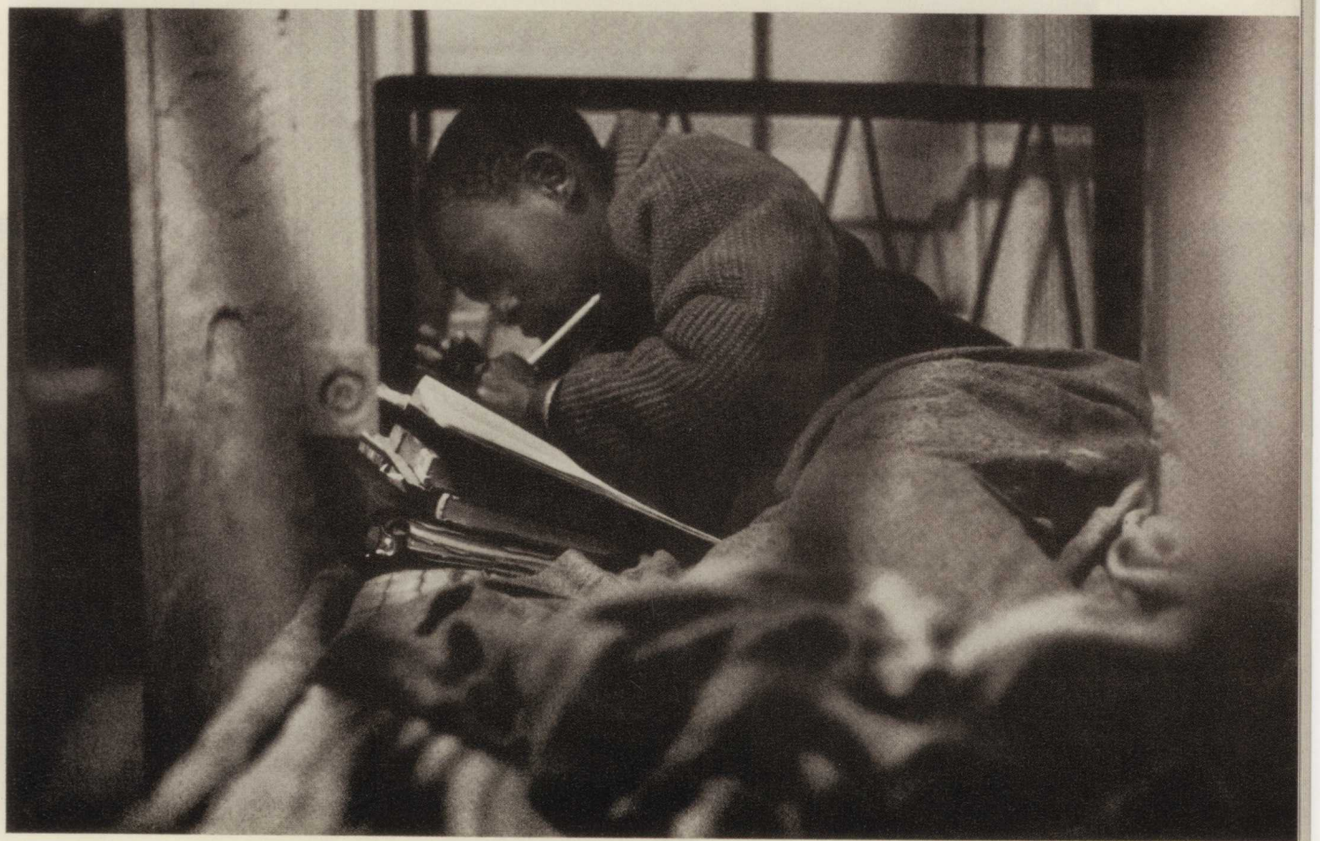
A HARLEM FAMILY 1967

Steidl/The Gordon Parks Foundation/The Studio Museum in Harlem



Norman Sr., barely able to read or write himself, tries helping Riel

For a few minutes the comics lure Riel and Kenneth



With a coat on and a blanket, Kenneth's fingers are still stiff

The books are not Ellen's but they may help her to another world

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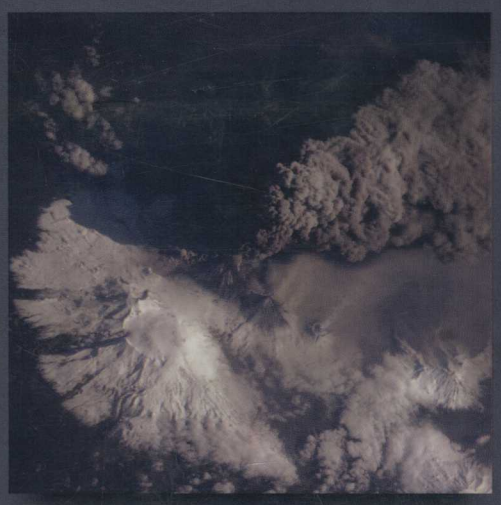

ORBIT

NASA ASTRONAUTS PHOTOGRAPH THE EARTH

"This book provides a very real sense of the wonder and majesty that astronauts experience as they gaze upon our world from above."

—SENATOR JOHN GLENN

The first American to orbit and photograph the Earth

"A magnificent testament to what we see and feel from space: Earth so splendid, Earth so fragile."

—FRANK BORMAN

Commander of Apollo 8, *the first mission to orbit the Moon*

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